

# Average Causal Effect in Observational Studies IV

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Spring 2026

## Implication of overlap assumption

### Theorem (D'Amour et al., 2021)

If  $\eta \leq e(\mathbf{X}_i) \leq 1 - \eta$  for some  $\eta \in (0, 1/2)$ , then

$$\begin{aligned} & \frac{1}{p} \sum_{k=1}^p |\mathbb{E}(X_{ik} \mid Z_i = 1) - \mathbb{E}(X_{ik} \mid Z_i = 0)| \\ & \leq \frac{1}{p^{1/2}} C^{1/2} \left[ e \lambda_{\max}^{1/2} \{\text{cov}(\mathbf{X}_i \mid Z_i = 1)\} + (1 - e) \lambda_{\max}^{1/2} \{\text{cov}(\mathbf{X}_i \mid Z_i = 0)\} \right] \end{aligned}$$

- $\lambda_{\max}^{1/2} \{\text{cov}(\mathbf{X}_i \mid Z_i = z)\}$  is usually smaller than  $O(p)$
- the right hand side converges to zero if the components of  $X$  are not highly correlated  $\rightsquigarrow$  means of  $X$  are nearly balanced

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  - ① Sharp RD Design: treatment assignment is based on a *deterministic* rule  $Z_i = \mathbf{1}(X_i \geq c)$
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$$e(X_i) = \begin{cases} 1 & \text{if } X \geq c \\ 0 & \text{if } X < c \end{cases}$$

- Originates from a study of the effect of scholarships on students' career plans (Thistlethwaite and Campbell. 1960. *J. of Educ. Psychol*)

# Regression discontinuity design

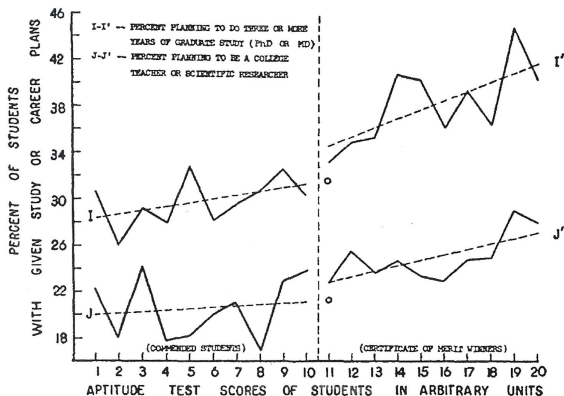


FIG. 3. Regression of study and career plans on exposure determiner.

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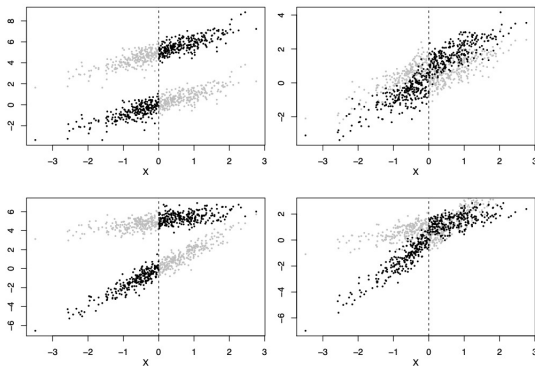
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- Advantage: internal validity
- Disadvantage: external validity

# Statistical inference: regression

- Graphical diagnostic sometimes shows the causal effect at the cutoff point



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- Linear regression models

$$\mathbb{E}(Y_i \mid Z_i = 1, X_i) = \alpha_1 + \gamma_1 X_i$$

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- Equivalent implementation

- coefficient of  $Z_i$  in regression of  $Y_i$  on  $\{1, Z_i, X_i - c, Z_i(X_i - c)\}$
- coefficient of  $Z_i$  in regression of  $Y_i$  on  $\{1, Z_i, (X_i - c)^+, (X_i - c)^-\}$ ,  
where  $(X_i - c)^+ = \max\{X_i - c, 0\}$  and  $(X_i - c)^- = \min\{X_i - c, 0\}$

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- *rdrobust* package in R

# Checking the identification assumption of RD design

- RD design does NOT require the **local randomization** or “as-if random” assumption within a window:

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- Examples of **sorting** or **manipulation of forcing variable**
  - reporting one's income just below the threshold to qualify for scholarships
  - students test score vs. teacher's assessment

# Continuity from nonrandom selection

- A simple model

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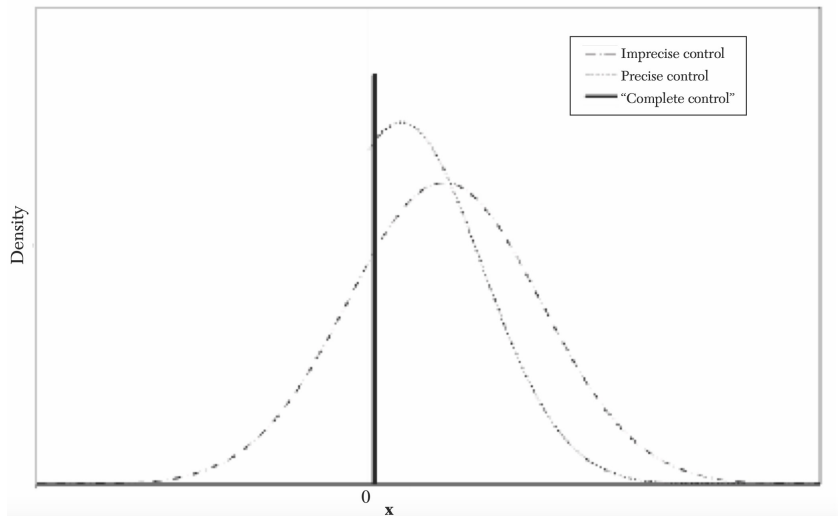
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- See Lee and Lemieux (2010) “Regression Discontinuity Designs in Economics”

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- Placebo test: check continuity for pre-treatment variables
  - expected not to have any effect
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- Check the continuity of the density of the forcing variable
- No unexplainable discontinuity in the outcome-forcing variable relationship at values other than at the cutoff value

## Incumbency advantage in the U.S. House (Lee, 2008)

- Re-election rate of the incumbent party is about 90% over the past 50 years
- Possible explanations
  - Representatives are using the privileges and resources of office to gain an “unfair” advantage over potential challengers.
  - Erikson (1971): incumbents, are, by definition, those politicians who were successful in the previous election. If what makes them successful is somewhat persistent over time, they should be expected to somewhat more successful when running for re-election.

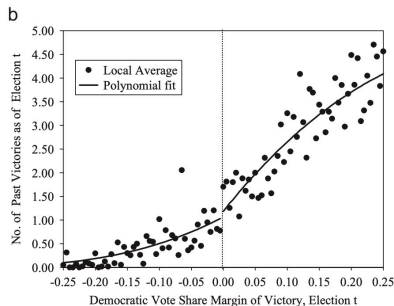
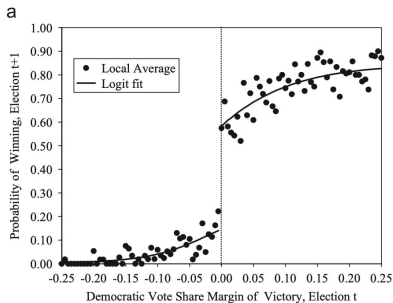
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- Question: is there any causal effect of being the incumbent party on the winning of the election
- Unit: congressional district
- Treatment  $Z_i$ : democrat is the incumbent party in a district
- Outcome  $Y_i$ : democrat winning the election



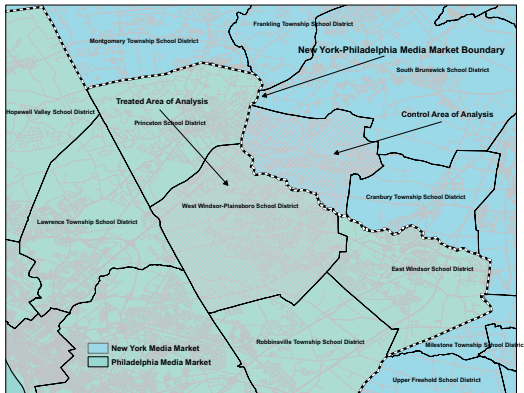
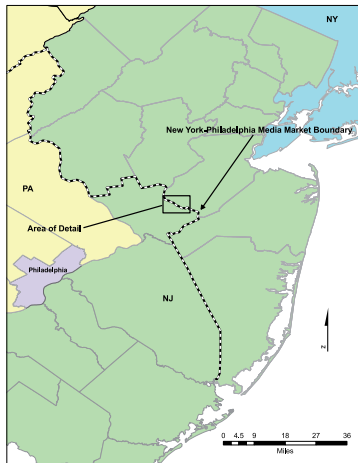
# RDD analysis of the incumbency advantage

- Forcing variable  $X_i$ : vote share for Democrat – vote share for republican
- Deterministic rule:  $Z_i = \mathbf{1}(X_i > 0)$
- Causal quantity:  $\mathbb{E}\{Y_i(1) - Y_i(0) \mid X_i = 0\}$



# Geographical RD Design (Keele and Titiunik. 2015. *Political Anal.*)

- RD in two dimensions
- Example: media markets



# Effect of sustained exposure to air pollution on life expectancy (Chen et al. 2013. *PNAS*)

- Chinese government established free winter heating of homes and offices via the provision of free coal for fuel boilers as a basic right.
- Due to budgetary limitations, this right was only extended to areas located in North China, which is defined by the line formed by the Huai River and Qinling Mountain range

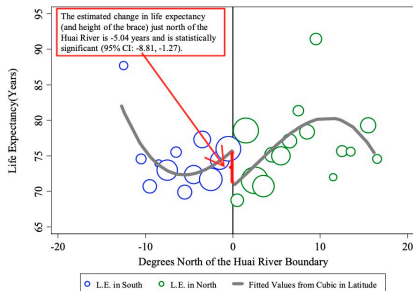
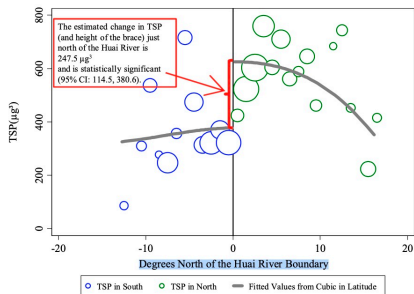


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- Their conclusion: the policy that greatly increases TSPs air pollution is causing the 500 million residents of Northern China to lose more than 2.5 billion life years of life expectancy



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- Model sensitivity
- Other issues
  - migration of population
  - people living in different parts of China differ a lot

# Summary

- Assessing overlap: plot propensity score for treatment and control group
- Regression discontinuity design
  - treatment is a deterministic function of covariates
  - Identify the ACE at the threshold
  - limited external validity  $\rightsquigarrow$  extrapolation required for generalization

# Suggested readings

- Regression discontinuity design
  - ANGRIST AND PISCHKE, Chapters 6.1
  - Cattaneo and Titiunik (2022) “Regression Discontinuity Designs”
- Statistical inference related
  - FAN AND GIJBELS “Local Polynomial Modelling and Its Applications”
  - bandwidth selection : Imbens and Kalyanaraman (2012) “Optimal bandwidth choice for the regression discontinuity estimator”
- Continuity versus local randomization
  - de la Cuesta and Imai (2016) “Misunderstandings about the Regression Discontinuity Design in Close Elections Dataverse”

## Suggested readings

- Interrupted time series
  - Miritrix (2022) “Using Simulation to Analyze Interrupted Time Series Designs”
- Non-extreme case: overlap violated for some values of  $X$ 
  - outcome modeling for extrapolation
  - overlap weights: Li and Zaslavsky (2018) “Balancing covariates via propensity score weighting”
  - stochastic intervention: Kennedy (2018) “Nonparametric Causal Effects Based on Incremental Propensity Score Interventions”